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Introduction:

The Museum Data Analysis case study is designed to provide insightful data-driven solutions for a hypothetical art museum database. The objective of this project is to extract, analyze, and interpret data to answer key business questions related to artworks, artists, museums, and painting styles. The dataset comprises multiple tables, including `work`, `museum`, `artist`, `subject`, `product_size`, `canvas_size`, `image_link`, and `museum_hours`, each containing relevant data on paintings, artists,

Q1: Fetch all the paintings which are not displayed on any museums?

```
Select *  
from work  
where museum_id is null
```

Q2: Are there museums without paintings?

```
Select *  
from museum m  
left join work w on m.museum_id= w.museum_id  
where work_id is null;
```

```
--OR  
select * from museum m  
where not exists  
    (select 1 from work w  
     where w.museum_id=m.museum_id  
    )
```

Q3: How many paintings have an asking price of more than their regular price?

```
Select *  
from product_size  
where sale_price > regular_price
```

Q4: Identify the paintings whose asking price is less than 50% of its regular price

```
Select *  
from product_size  
where sale_price < (regular_price/2)
```

Q5: Which canvas size costs the most?

```
Select cs.label, sale_price  
from product_size ps  
join canvas_size cs on cast(cs.size_id as TEXT) = ps.size_id  
order by 2 desc ,2 desc  
limit 1;
```

Q6: Delete duplicate records from work, product_size, subject and image_link tables

```
WITH duplicates AS (  
  SELECT  
    work_id, -- Assume 'id' is the primary key or unique identifier  
    ROW_NUMBER() OVER (PARTITION BY work_id, size_id ORDER BY work_id) AS rn  
  FROM product_size  
)  
delete from subject  
where work_id in (  
  select work_id from duplicates  
  where rn>1)
```

-----IMAGE LINK

```
WITH duplicates AS (  
  SELECT  
    work_id, -- Assume 'id' is the primary key or unique identifier  
    ROW_NUMBER() OVER (PARTITION BY work_id ORDER BY work_id) AS rn  
  FROM image_link  
)  
  
delete from image_link  
where work_id in (select work_id from duplicates  
  where rn>1)
```

Q7: Identify the museums with invalid city information in the given dataset

```
Select city
from museum
where city ~ '[0123456789]'
```

OR –MYSQL

```
Select city
from museum
where city regex '[0123456789]'
```

Q8: Museum_Hours table has 1 invalid entry. Identify it and remove it.

```
delete from museum_hours
      where ctid not in (select min(ctid)
                        from museum_hours
                        group by museum_id, day );
```

Q9: Fetch the top 10 most famous painting subject

```
select s.subject,count(w.work_id) as no_of_paintings
from work w
join subject s on s.work_id=w.work_id
group by s.subject
order by 2 desc
limit 10;
```

OR –Rank help with the ties.

```
select *
from (
      select s.subject,count(1) as no_of_paintings
      ,rank() over(order by count(1) desc) as ranking
      from work w
      join subject s on s.work_id=w.work_id
      group by s.subject
    )x
```

where ranking <= 10;

Q10: Identify the museums which are open on both Sunday and Monday. Display museum name, city.

```
Select distinct mh1.museum_id,  
m.name, count(day)  
from museum_hours mh1  
join museum m on mh1.museum_id= m.museum_id  
where day in ('Monday', 'Sunday')  
group by 1,2  
having count(day)=2  
order by 1;
```

--OR

```
Select distinct mh1.museum_id  
from museum_hours mh1  
where day ='Sunday'  
and exists (  
select 1 from museum_hours mh2  
where mh2.museum_id= mh1.museum_id  
and mh2.day='Monday'  
)  
order by 1;
```

Q11: How many museums are open every single day?

```
Select count(*)  
from (  
    Select museum_id, count(day)  
    from museum_hours  
    group by 1  
    having count(day)= 7) x
```

Q12: Which are the top 5 most popular museum? (Popularity is defined based on most no of paintings in a museum)

–Without Rank

```
select count(w.work_id) as work_count, m.name,m.country,m.name
from work w
join museum m on w.museum_id= m.museum_id
where w.museum_id is not null
group by 2,3,4
order by 1 desc
limit 5;
```

OR – With rank

```
select m.name as museum, m.city,m.country,x.no_of_painintgs
      from ( select m.museum_id, count(work_id) as no_of_painintgs
              ,rank()over(order by count(w.work_id) desc) rk
              from work w
              join museum m on m.museum_id=w.museum_id
              group by m.museum_id) x
join museum m on m.museum_id=x.museum_id
where x.rk<=5
order by 4 desc;
```

Q13: Who are the top 5 most popular artist? (Popularity is defined based on most no of paintings done by an artist)

```
select count(w.work_id) as work_count, a.full_name, a.nationality
from work w
join artist a on w.artist_id= a.artist_id
group by 2,3
order by 1 desc
limit 5;
```

OR With Rank

```
select work_count, x.full_name, x.nationality
from
      (select count(w.work_id) as work_count, a.full_name, a.nationality,
              rank() over (order by count(work_id)desc ) rk
              from work w
              join artist a on w.artist_id= a.artist_id
              group by 2,3
              order by 1 desc
              ) X
```

where x.rk <= 5

Q14: Display the 3 least popular canva sizes

```
select x.label, x.cnt as painting_count, x.rk as ranking
from
    (
        Select count(w.work_id) cnt, cs.size_id, cs.label,
        dense_rank() over (order by count(w.work_id)) rk
        from work w
        join product_size ps on ps.work_id=w.work_id
        join canvas_size cs on ps.size_id = cast(cs.size_id as TEXT)
        GROUP by 2,3
    )
x
where x.rk <=3
```

Q15 : Which museum is open for the longest during a day. Dispay museum name, state and hours open and which day?

Without Limit→

```
Select *
from (
    Select distinct mh.museum_id,m.name,
    m.state,
    close, open,
    cast(replace(close,' ','') as time) - cast (open as time) as open_hours,
    rank() over (order by (cast(replace(close,' ','') as time) - cast (open as time))desc) rk
    from museum_hours mh
    join museum m on mh.museum_id= m.museum_id
    order by open_hours desc
) X
where x.rk= 1;
```

OR

```
Select distinct mh.museum_id,m.name,
m.state,
```

```
close, open,  
cast(replace(close,' ','') as time) - cast (open as time) as open_hours  
from museum_hours mh  
join museum m on mh.museum_id= m.museum_id  
order by open_hours desc  
limit 1;
```

Q16: Which museum has the most no of most popular painting style?

```
select x.no_paintings, x.style,x.name  
from  
    (Select count(w.work_id) as no_paintings  
    ,style, m.name,  
    dense_rank() over(order by count(w.work_id)desc) rk  
    from work w  
    join museum m on m.museum_id= w.museum_id  
    group by 2,3 ) X  
where x.rk= 1  
and style is not null  
order by 1 desc  
limit 1;
```

Q17: Identify the artists whose paintings are displayed in multiple countries

```
Select a.full_name, count(distinct m.country) no_countries  
from work w  
join museum m on w.museum_id = m.museum_id  
join artist a on a.artist_id=w.artist_id  
group by 1  
having count(distinct m.country) >1  
order by 2 desc
```

Q18 : Display the country and the city with most no of museums. Output 2 seperate columns to mention the city and country. If there are multiple value, seperate them with comma.

```

with cte1 as
(
  Select count(museum_id) cn1, city,
  rank() over(order by (count(museum_id))desc) rk1
  from museum
  group by 2
),
cte2 as
(
  Select count(museum_id) cn2, country,
  rank() over(order by (count(museum_id))desc) rk2
  from museum
  group by 2
)
select
string_agg (distinct cte2.country,' , ' ) as Countries,
string_agg (cte1.city,' , ' ) Cities

from cte1
cross join cte2
-- or 1=1 (always true)
where cte1.rk1= 1 and cte2.rk2=1

```

Q19: Identify the artist and the museum where the most expensive and least expensive painting is placed. Display the artist name, sale_price, painting name, museum name, museum city and canvas label

```

WITH cte AS (
  SELECT
    ps.*,
    RANK() OVER (ORDER BY ps.sale_price DESC) AS rk,
    RANK() OVER (ORDER BY ps.sale_price ASC) AS rk_asc
  FROM product_size ps
)
SELECT
  w.name as painting
  , cte.sale_price

```



```

, a.full_name as artist
, m.name as museum, m.city
, cz.label as canvas
FROM cte
join work w on w.work_id=cte.work_id
join museum m on m.museum_id=w.museum_id
join artist a on a.artist_id=w.artist_id
join canvas_size cz on cast(cz.size_id as TEXT) = cte.size_id
WHERE rk = 1 OR rk_asc = 1;

```

Q20: Which country has the 5th highest no of paintings?

```

Select count(w.work_id), m.country
from work w
join museum m on w.museum_id = m.museum_id
group by 2
order by 1 desc
limit 1 offset 4;

```

OR - without limit

```

select x.painting_count, x.country
from
    (Select count(w.work_id) Painting_count, m.country,
    rank() over( order by(count(w.work_id) ) desc ) rk
    from work w
    join museum m on w.museum_id = m.museum_id
    group by 2 ) X
where x.rk= 5

```

Q21: Which are the 3 most popular and 3 least popular painting styles?

```

(select style, count(work_id) as no_of_paintings , 'Most Popular' as Popularity
from work
where style is not null
group by style
order by 2 desc
limit 3 )

```

union

```
(select style, count(work_id) as no_of_paintings , 'Least Popular' as Popularity
from work
where style is not null
group by style
order by 2 ASC
limit 3)
```

OR

```
WITH x AS (
  SELECT
    COUNT(w.work_id) AS no_paintings,
    w.style,
    DENSE_RANK() OVER (ORDER BY COUNT(w.work_id) DESC) AS rk,
    DENSE_RANK() OVER (ORDER BY COUNT(w.work_id)) AS rkn
  FROM work w
    where style is not null
  GROUP BY w.style
)
SELECT x.no_paintings, x.style,
case
  when rk <=3 then 'Most Popular'
  else 'Least Popular'
end as remarks
FROM x
WHERE rk <= 3 OR rkn <= 3;
```

with cte as

```
(select style, count(1) as cnt
, rank() over(order by count(1) desc) rnk
, count(1) over() as no_of_records
from work
where style is not null
group by style)
select style
, case when rnk <=3 then 'Most Popular' else 'Least Popular' end as remarks
from cte
where rnk <=3
```

or $\text{rnk} > \text{no_of_records} - 3$;

Q22: Which artist has the most no of Portraits paintings outside USA?. Display artist name, no of paintings and the artist nationality.

```
select x.full_name,x.nationality, x.no_of_painting
from (
    select count(w.work_id) as no_of_painting, country, w.artist_id, a.full_name, a.nationality,
    rank() over(partition by w.artist_id order by count(w.work_id) desc ) rk
    from work w
    join subject s on w.work_id= s.work_id
    join museum m on w.museum_id= m.museum_id
    join artist a on a.artist_id= w.artist_id
    where country != 'USA'
    and s.subject= 'Portraits'
    group by 2 , 3,4,5
    order by 1 desc ) X
where x.rnk= 1
limit 1;
```